



MySQL Assignment: E-Commerce Database

Objective

Create a MySQL database for an **E-Commerce Full Stack Project**.

Students will later use this database in:

Node.js, Express.js, React.js, MySQL, and Sequelize.

Database Name

ecommerce_db

Tables to Create

1. Users Table

Create a table named `users` with:

Column	Requirement
id	Primary key, auto increment
name	User full name
email	Unique email
password	User password
role	user / admin

created_at	Default current timestamp
------------	---------------------------

2. Categories Table

Create a table named `categories` with:

Column	Requirement
id	Primary key, auto increment
name	Category name
description	Category description
created_at	Default current timestamp

3. Products Table

Create a table named `products` with:

Column	Requirement
id	Primary key, auto increment
title	Product title
description	Product description

price	Product price
stock	Available quantity
image	Product image name/path
category_id	Foreign key from categories table
created_at	Default current timestamp

4. Cart Table

Create a table named `cart` with:

Column	Requirement
id	Primary key, auto increment
user_id	Foreign key from users table
product_id	Foreign key from products table
quantity	Product quantity
created_at	Default current timestamp

Assignment Tasks

Task 1: Database Setup

Create a database named:

`ecommerce_db`

Use this database for all tables.

Task 2: Create Tables

Create the following tables:

1. `users`
2. `categories`
3. `products`
4. `cart`

Use proper:

- Primary Key
- Auto Increment
- Unique Key
- Foreign Key
- Default Timestamp

Task 3: Insert User Data

Insert minimum **5 users**.

Data should include:

- Normal users
 - At least 1 admin user
-

Task 4: Insert Category Data

Insert minimum **5 categories**

Insert minimum 5 categories.

Example categories:

- Electronics
 - Fashion
 - Books
 - Home Appliances
 - Beauty
-

Task 5: Insert Product Data

Insert minimum **10 products**.

Each product must have:

- Title
 - Description
 - Price
 - Stock
 - Image
 - Category ID
-

Task 6: Insert Cart Data

Insert minimum **5 cart records**.

Each cart record should contain:

- User ID
 - Product ID
 - Quantity
-

Task 7: Fetch All Records

Write queries to fetch:

- All users
 - All categories
 - All products
 - All cart records
-

Task 8: Product Filtering

Write queries to find:

- Products below ₹5000
 - Products between ₹1000 and ₹30000
 - Products having stock greater than 10
 - Products having stock less than 5
-

Task 9: Search Product

Write a query to search product by title.

Example search:

mobile

Task 10: Category Wise Products

Write a query to fetch products by category.

Example:

Fetch all products from Electronics category.

Join Query Tasks

Task 11: Product With Category Name

Write a join query to show:

- Product ID
 - Product title
 - Price
 - Stock
 - Category name
-

Task 12: Cart Details

Write a join query to show:

- Cart ID
 - User name
 - Product name
 - Product price
 - Quantity
 - Total price
-

Update Tasks

Task 13: Update Product Price

Update price of any one product.

Task 14: Update Product Stock

Update stock of any product.

Example:

Reduce stock after purchase.

Task 15: Update User Role

Change one user role from `user` to `admin`.

Task 16: Remove Product From Cart

Delete one cart item.

Task 17: Delete Product

Delete one product from products table.

Task 18: Product Count

Write queries to find:

- Total number of products
 - Total number of users
 - Total number of categories
-

Task 19: Category Wise Product Count

Write a query to count products in each category.

Task 20: Price Analysis

Write queries to find:

- Highest price product
- Lowest price product
- Average product price

Task 21: Cart Total

Write a query to calculate total cart amount for one user.

Task 22: Inventory Value

Write a query to calculate total inventory value.

Formula:

$\text{price} \times \text{stock}$

Task 23: Products Never Added to Cart

Write a query to find products that have never been added to any cart.

Expected Output

Product ID	Product Name
5	Keyboard
8	Mouse

Task 24: User With Highest Cart Value

Write a query to find the user whose cart amount is highest.

Expected Output

User Name	Total Amount
Rahul	75000

Task 25: Category Revenue Report

Calculate potential revenue category-wise.

Formula:

$\text{price} \times \text{stock}$

Expected Output

Category	Inventory Value
Electronics	500000
Fashion	80000

Task 26: Top 3 Most Expensive Products

Write a query to display the top 3 most expensive products.

Expected Output

Product	Price
---------	-------

Laptop 55000
Mobile 25000
Monitor 18000

Task 27: Product Statistics Report

Generate a report showing:

- Total Products
 - Average Product Price
 - Highest Product Price
 - Lowest Product Price
 - Total Stock Available
 - Out of Stock Products Count
-

Task 28: Category Performance Report

Display:

- Category Name
 - Number of Products
 - Average Product Price
 - Total Stock
-

Task 29: User Cart Summary

Display:

- User Name
 - Number of Products in Cart
 - Total Quantity
 - Total Cart Amount
-

Task 30: Complete E-Commerce Database Report

Create a report containing:

- Total Users
- Total Categories
- Total Products
- Total Cart Records
- Total Inventory Value
- Highest Priced Product
- Lowest Priced Product
- Most Added Product in Cart
- Out of Stock Products
- Average Product Price